MTU Cork

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ELA: English Language Analysis Pipeline

An AI-Powered Hierarchical Linguistic Contract System for Text, Audio & Video

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Introduction

The ELA (English Language Assistant) system investigates how hybrid NLP and Generative AI can support bilingual comprehension of authentic English materials without simplification. ELA transforms real-world media — text, audio, and video — into structured Recursive Linguistic Elements (RLE): hierarchical JSON representations at Sentence → Phrase → Word granularity, linking syntax, translation, CEFR level, and contextual notes. The system integrates deterministic NLP (spaCy, CoreNLP) with ML-based annotation to enable cognitive scaffolding for language learners. Deployed as an Android-installable Progressive Web App (PWA) and a Tauri 2 desktop application (Linux .deb), both running fully client-side with no cloud dependency.

NLP Core Pipeline

The parsing engine uses spaCy 3.7 for deterministic sentence segmentation, dependency parsing, and morphological analysis. A rule-based Tense-Aspect-Mood (TAM) engine extracts grammatical tense (present/past/future), aspect (simple/progressive/perfect), mood (indicative/subjunctive/imperative), voice (active/passive), and finiteness from each clause.

A phrase pattern matcher groups words into linguistically motivated phrases (NP, VP, PP, ADJP), forming the three-level hierarchy: Sentence → Phrase → Word. All structural annotations are validated against a dual-schema contract (v1 and v2_strict) before persisting to the local or shared database.

Pipeline Flow

① Input Text / Media

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② spaCy Parser

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③ TAM Rules Engine

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④ Phrase Builder

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⑤ Contract Validator

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⑥ JSON Contract Output

Machine Learning Components

Three ML components enrich the base linguistic contract:

(1) CEFR Classification — a Tabular ML ensemble (XGBoost + Random Forest + Logistic Regression) trained on a 3,000-sentence CEFR-balanced corpus predicts English proficiency level (A1–C2) per sentence and word.

(2) Linguistic Note Generation — a fine-tuned T5 (Text-to-Text Transfer Transformer) model generates pedagogical notes for grammar structures, backed by deterministic rule-based template fallbacks.

(3) Speech Recognition — OpenAI Whisper (base/medium) performs client-side Automatic Speech Recognition (ASR) for audio and video inputs, with FFmpeg handling media pre-processing.

Performance Metrics

78.3%

CEFR Classification Accuracy

XGBoost ensemble on 3,000-sentence corpus

0.81

Note Relevance (ROUGE-L)

T5 fine-tuned model evaluation

8.2%

ASR Word Error Rate

Whisper base, clean speech test set

Conclusions

This project demonstrates that explainable, AI-assisted annotation can make authentic English input pedagogically accessible without simplification. ELA successfully produces Recursive Linguistic Elements (RLE) from multimodal inputs (text, PDF, audio, video) with CEFR classification accuracy of 78.3%, validating a hybrid rule-based NLP and lightweight ML approach for bilingual language learning. The system confirms that cognitive scaffolding via structured RLE and bilingual visualisation meaningfully reduces the barrier to authentic content comprehension.

Future directions include: expanding to additional target languages beyond English→Russian translation, integrating larger transformer models (BERT, RoBERTa) for improved CEFR prediction, adding real-time streaming analysis for live audio, and developing a teacher-facing analytics dashboard for classroom deployment.

References

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